



## Reflect with us - Week 5



This assignment will not be graded and is only for practice.

1 point

Consider a map  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  defined as  $T(v) = Av$ , where  $v = \begin{bmatrix} x \\ y \end{bmatrix}$ ,  $v = [xy]$ ,

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  and  $\det(A) \neq 0$ . Which of the following options are correct?

☐

$T$  is a linear transformation.

☐

$T$  is both one-one and onto.

☐

$T$  is neither one-one nor onto.

☐

$T$  is one-one but not onto.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T$  is a linear transformation.

$T$  is both one-one and onto.

- Step 1:Step 1:

1 point

What is  $T(v_1 + v_2)$ ?

☐

$A(v_1 + v_2)$ .

☐

$Av_1 + Av_2$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$A(v_1 + v_2)$ .

Recall:  $A(v_1 + v_2) = Av_1 + Av_2$

- Step 2:Step 2:

1 point

Is  $T(v_1 + v_2) = T(v_1) + T(v_2)$ ?

☐

Yes.

☐

No.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Yes.

Check:  $T(cv) = cT(v)$

- Hence  $T$  is linear transformation. So Option 1 is true.

- Another method: Write the definition of  $T$  explicitly as,

$$T(x, y) = (ax + by, cx + dy)$$



- Step 1: Step 1: What is  $T((x_1, y_1) + (x_2, y_2))T((x_1, y_1) + (x_2, y_2))$ ?

$$\begin{aligned} T(x_1 + x_2, y_1 + y_2) &= (a(x_1 + x_2) + b(y_1 + y_2), c(x_1 + x_2) + d(y_1 + y_2)) \\ &= (ax_1 + ax_2 + by_1 + by_2, cx_1 + cx_2 + dy_1 + dy_2) \\ &= (ax_1 + by_1, cx_1 + dy_1) + (ax_2 + by_2, cx_2 + dy_2) = T(x_1, y_1) + T(x_2, y_2) \end{aligned}$$

$$(a(x_1 + x_2) + b(y_1 + y_2), c(x_1 + x_2) + d(y_1 + y_2)) = (ax_1 + ax_2 + by_1 + by_2, cx_1 + cx_2 + dy_1 + dy_2) = (ax_1 + by_1, cx_1 + dy_1) + (ax_2 + by_2, cx_2 + dy_2) = T(x_1, y_1) + T(x_2, y_2)$$

- Step 2: Step 2:

1 point

Is  $T(c(x, y)) = cT(x, y)T(c(x, y)) = cT(x, y)$ ?

☐ Yes.

☐ No.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Yes.

- Hence  $TT$  is linear transformation. So Option 1 is true.

Now in the given problem  $T(v) = 0T(v) = 0$ , implies  $Av = 0Av = 0$ . This is the matrix representation of the system of linear equations:

$$\begin{aligned} ax + by &= 0 \\ cx + dy &= 0 \end{aligned} \quad ax + bycx + dy = 0 = 0$$

- Step 3: Step 3:

1 point

If the system of linear equations  $Av = 0Av = 0$  has a unique solution, then what is the solution?

☐

$v = 0v = 0$

☐

$v$  can be any vector in  $R^2$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$v = 0v = 0$

- Step 4: Step 4:

1 point

What is the condition on  $AA$ , for which the system of linear equations  $Av = 0Av = 0$  has a unique solution? [Recall from Weeks 1 and 2]

☐

$\det(A) = 0 \det(A) = 0$ .

☐

$\det(A) \neq 0 \det(A) = 0$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\det(A) \neq 0 \det(A) = 0$ .

Concluding, we can say that, if  $\det(A)$  is non-zero, then  $T(v) = 0T(v) = 0$  implies  $v = 0$ . Hence  $TT$  is one to one.

- Let  $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \in \mathbb{R}^2$   $w = [w_1 w_2] \in \mathbb{R}^2$ . We are going to find whether there exists a vector  $v = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$   $v = [xy] \in \mathbb{R}^2$ , such that  $T(v) = Av = w$ .

$Av = w$  is the matrix representation of the system of linear equation:

$$\begin{aligned} ax + by &= w_1 \\ cx + dy &= w_2 \end{aligned}$$

- Step 5: Step 5:

1 point

If  $\det(A) \neq 0$ , then what can we say about the solution of the above system of linear equations?

- ☐ No solution.  
☐ Unique solution.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Unique solution.

Concluding from this, we can say that, if  $\det(A) \neq 0$ , then  $T$  is onto.

Hence, Options 1 and 2 are the correct options. Hence, Options 1 and 2 are the correct options.

[Check Answers](#)

Your score is: 0/7